

Chapter 5 Integration

Antidifferentiation

Given a function $f(x)$, an antiderivative of $f(x)$ is a function $F(x)$ such that

$$F'(x) = f(x).$$

We represent the family of antiderivatives by $F(x) + C$, where $F(x)$ is any antiderivative of $f(x)$ and C is a constant.

The family of antiderivatives of $f(x)$ is also denoted by the indefinite integral symbol

$$\int f(x) dx$$

Power Rule $\int x^r dx = \frac{1}{r+1} x^{r+1} + C, r \neq -1$ $\frac{d}{dx} \left(\frac{1}{r+1} x^{r+1} \right) = x^r$

Log Rule $\int \frac{1}{x} dx = \ln|x| + C, x \neq 0$ $\frac{d}{dx} (\ln|x|) = \frac{1}{x}$

Exponential Rule $\int e^{kx} dx = \frac{1}{k} e^{kx} + C, k \neq 0.$ $\frac{d}{dx} \left(\frac{1}{k} e^{kx} \right) = e^{kx}$

Sum Rule $\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx$

Constant Multiple Rule $\int kf(x) dx = k \int f(x) dx.$

EXAMPLE 1 A Power Function

Find an antiderivative of $f(x) = x^2$.

SOLUTION The derivative of x^3 is $3x^2$, which is almost the same as x^2 except for a factor of 3. To make this factor 1 instead of 3, consider the function $\frac{1}{3}x^3$. Then

$$\begin{aligned} \frac{d}{dx} \left(\frac{1}{3} x^3 \right) &= \frac{1}{3} \left(\frac{d}{dx} x^3 \right) \quad \text{Constant multiple rule for differentiation.} \\ &= \frac{1}{3} \cdot 3x^2 = x^2 \quad \text{Differentiate } x^3 \text{ then simplify.} \end{aligned}$$

So, $\frac{1}{3}x^3$ is an antiderivative of x^2 . Is this the only antiderivative of x^2 ? Consider the function $\frac{1}{3}x^3 + 5$. Since

$$\begin{aligned} \frac{d}{dx} \left(\frac{1}{3} x^3 + 5 \right) &= \frac{d}{dx} \left(\frac{1}{3} x^3 \right) + \frac{d}{dx} (5) && \text{Sum rule for differentiation.} \\ &= x^2 + 0 && \text{Derivative of a constant is 0} \\ &= x^2 \end{aligned}$$

Since the derivative of $\frac{1}{3}x^3 + 5$ is x^2 , we conclude that $\frac{1}{3}x^3 + 5$ is also an antiderivative of x^2 . More generally, if C is any constant, then $F(x) = \frac{1}{3}x^3 + C$ is an antiderivative of x^2 , because

$$\begin{aligned} F'(x) &= \frac{d}{dx} \left(\frac{1}{3}x^3 + C \right) \\ &= \frac{d}{dx} \left(\frac{1}{3}x^3 \right) + \frac{d}{dx} (C) \\ &= x^2 + 0 \\ &= x^2. \end{aligned}$$

$$\int x^2 dx = \frac{1}{3}x^3 + C$$

Sum rule for differentiation.

Derivative of a constant is 0.

EXAMPLE 2 An Exponential Function

Find an antiderivative of $f(x) = e^{-2x}$.

$$\int e^{kx} dx = \frac{e^{kx}}{k} + C$$

SOLUTION

Recall that the derivative of e^{rx} is just a constant times e^{rx} . For an antiderivative of e^{-2x} , try a function of the form ke^{-2x} , where k is some constant to be determined. Then,

$$\begin{aligned} \frac{d}{dx}(ke^{-2x}) &= k \cdot \left(\frac{d}{dx} e^{-2x} \right) && \text{Constant multiple rule for differentiation.} \\ &= -2ke^{-2x} && \text{Derivative of } e^{-2x} \text{ is } -2e^{-2x} \end{aligned}$$

We want this derivative to be e^{-2x} , so choose k to make $-2k = 1$; that is, choose $k = -\frac{1}{2}$. Then,

$$\frac{d}{dx} \left(-\frac{1}{2}e^{-2x} \right) = \left(-\frac{1}{2} \right) (-2e^{-2x}) = 1 \cdot e^{-2x} = e^{-2x}.$$

Thus, $-\frac{1}{2}e^{-2x}$ is an antiderivative of e^{-2x} . Also, for any constant C , the function $F(x) = -\frac{1}{2}e^{-2x} + C$ is an antiderivative of e^{-2x} , because

$$\frac{d}{dx} \left(-\frac{1}{2}e^{-2x} + C \right) = e^{-2x} + 0 = e^{-2x}$$

$$\int e^{-2x} dx = \frac{e^{-2x}}{-2} + C$$

EXAMPLE 3 Using the Power Rule

Determine

(a) $\int \sqrt{x} dx$

(b) $\int \frac{1}{x^2} dx$

SOLUTION Both integrals follow by applying the power rule.

(a) Take $r = \frac{1}{2}$; then

$$\int x^{\frac{1}{2}} dx = \frac{1}{\frac{1}{2} + 1} x^{\frac{1}{2} + 1} + C = \frac{1}{\frac{3}{2}} x^{\frac{3}{2}} + C = \frac{2}{3} x^{\frac{3}{2}} + C.$$

$$\int x^r dx = \frac{x^{r+1}}{r+1} + C$$

check:

$$\begin{aligned} \frac{d}{dx} \left(\frac{2}{3} x^{\frac{3}{2}} + C \right) &= \left(\frac{2}{3} \right) \left(\frac{3}{2} \right) x^{\frac{3}{2} - 1} + 0 = x^{\frac{1}{2}} \end{aligned}$$

(b) Take $r = -2$, then,

$$\int x^{-2} dx = \frac{1}{-2+1} x^{-2+1} + C = -x^{-1} + C = -\frac{1}{x} + C.$$

EXAMPLE 4 Using Basic Properties

Compute $\int \left(x^{-3} + 7e^{5x} + \frac{4}{x} \right) dx$.

SOLUTION

$$\begin{aligned} & \int x^{-3} + 7e^{5x} + \frac{4}{x} dx \\ &= \int x^{-3} dx + \int 7e^{5x} dx + \int \frac{4}{x} dx \\ &= \frac{x^{-3+1}}{-3+1} + 7 \frac{e^{5x}}{5} + 4 \ln|x| + C \\ &= -\frac{1}{2} x^{-2} + \frac{7}{5} e^{5x} + 4 \ln|x| + C \quad \# \end{aligned}$$

The Definite Integral

Suppose that $f(x)$ is a continuous function on $a \leq x \leq b$. The definite integral of $f(x)$ from a to b is the number

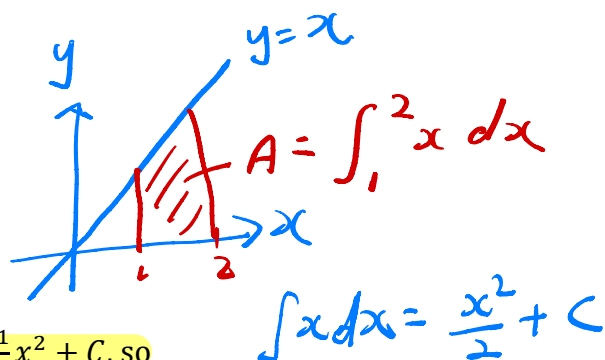
$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\int f(x) dx = F(x)$$

where F is any antiderivative of f . The number $F(b) - F(a)$ is abbreviated by $F(x)|_a^b$.

$$= F(x) \Big|_a^b = F(b) - F(a)$$

$$\begin{aligned}
 \int_a^b (f(x) + g(x))dx &= \int_a^b f(x)dx + \int_a^b g(x)dx \\
 \int_a^b kf(x)dx &= k \int_a^b f(x)dx \\
 \int_a^b f(x)dx &= - \int_b^a f(x)dx \\
 \int_a^a f(x)dx &= 0 \\
 \int_a^c f(x)dx + \int_c^b f(x)dx &= \int_a^b f(x)dx.
 \end{aligned}$$

EXAMPLE 5 Computing a Definite IntegralEvaluate $\int_1^2 x dx$.SOLUTION An antiderivative of $f(x) = x$ is $F(x) = \frac{1}{2}x^2 + C$, so

$$\int_1^2 x dx = F(2) - F(1) = \left(\frac{1}{2}(2)^2 + C\right) - \left(\frac{1}{2}(1)^2 + C\right) = 2 - \frac{1}{2} = \frac{3}{2}.$$

$$* \int_1^2 x dx = \left. \frac{x^2}{2} \right|_1^2 = \frac{2^2}{2} - \frac{1^2}{2} = 2 - \frac{1}{2} = \frac{3}{2}$$

EXAMPLE 6 Definite Integral of an ExponentialEvaluate $\int_{-1}^1 e^{-t} dt$.

$$\int e^{kx} dx = \frac{e^{kx}}{k} + C$$

SOLUTION To find an antiderivative of $f(t) = e^{-t}$, we use the exponential rule of the previous section, with $k = -1$, set the arbitrary constant $C = 0$, and get $F(t) = -e^{-t}$. Thus,

$$\int_{-1}^1 e^{-t} dt = -e^{-t} \Big|_{-1}^1 = -e^{-1} - (-e^{-(-1)}) = e - e^{-1} = e - \frac{1}{e} \approx 2.35$$

$$\int e^{-t} dt = \frac{e^{-t}}{-1} + C = -e^{-t}$$

Correct it to
2 decimal places

EXAMPLE 7 Using Properties of the IntegralEvaluate $\int_1^3 (6x - 18x^3)dx - 2\int_1^3 (x - x^3)dx$.**SOLUTION** Integrals over the same interval can be combined by the use of . We have

$$\begin{aligned}
& \int_1^3 (6x - 18x^3)dx - 2\int_1^3 (x - x^3)dx \\
&= \int_1^3 (6x - 18x^3)dx + \int_1^3 (-2x + 2x^3)dx \\
&= \int_1^3 ((6x - 18x^3) + (-2x + 2x^3))dx \\
&= \int_1^3 (4x - 16x^3)dx.
\end{aligned}$$

$\int 4x dx = \frac{4x^2}{2} = 2x^2$
// f(x)

An antiderivative of $f(x) = 4x - 16x^3$ is $F(x) = \frac{4}{2}x^2 - \frac{16}{4}x^4 = 2x^2 - 4x^4$. Thus,

$$\begin{aligned}
\int_1^3 (4x - 16x^3)dx &= F(3) - F(1) \\
&= (2(3^2) - 4(3^4)) - (2(1)^2 - 4(1)^4) \\
&= (18 - 324) - (2 - 4) = -306 - (-2) = -304
\end{aligned}$$

$$\begin{aligned}
\int 16x^3 dx &= \frac{16x^4}{4} \\
&= 4x^4
\end{aligned}$$

EXAMPLE 8Evaluate the definite integral $\int_{\frac{1}{2}}^1 \left(7x + \frac{1}{x}\right) dx$.

$$\begin{aligned}
& \int_1^3 (4x - 16x^3) dx = (2x^2 - 4x^4) \Big|_1^3 = [2(3)^2 - 4(3)^4] - \\
& \quad [2(1)^2 - 4(1)^4] = -304
\end{aligned}$$

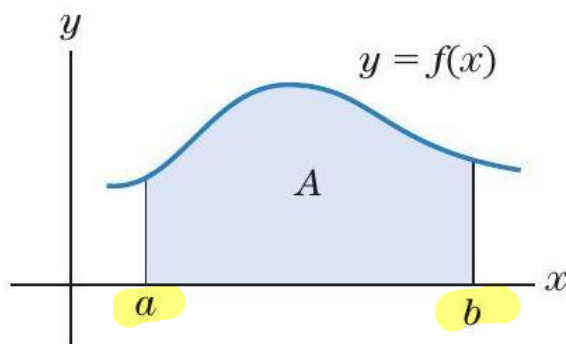
$$\begin{aligned}
\int_{\frac{1}{2}}^1 \left(7x + \frac{1}{x}\right) dx &= \left(7 \frac{x^{1+1}}{1+1} + \ln|x|\right) \Big|_{\frac{1}{2}}^1 \\
&= \left(\frac{7}{2}x^2 + \ln|x|\right) \Big|_{\frac{1}{2}}^1 \\
&= \left(\frac{7}{2}(1)^2 + \ln|1|\right) - \left(\frac{7}{2}\left(\frac{1}{2}\right)^2 + \ln\left|\frac{1}{2}\right|\right) \\
&= \left(\frac{7}{2} + 0\right) - \left[\left(\frac{7}{2}\right)\left(\frac{1}{4}\right) + \ln\left|\frac{1}{2}\right|\right] \\
&\approx \underline{\underline{3.32}}
\end{aligned}$$

direct
checked
by
calculator

The Definite Integral and Area under a Graph

Suppose that $f(x)$ is a continuous nonnegative function on $a \leq x \leq b$. The area of the region is given by

$$A = \int_a^b f(x) dx$$

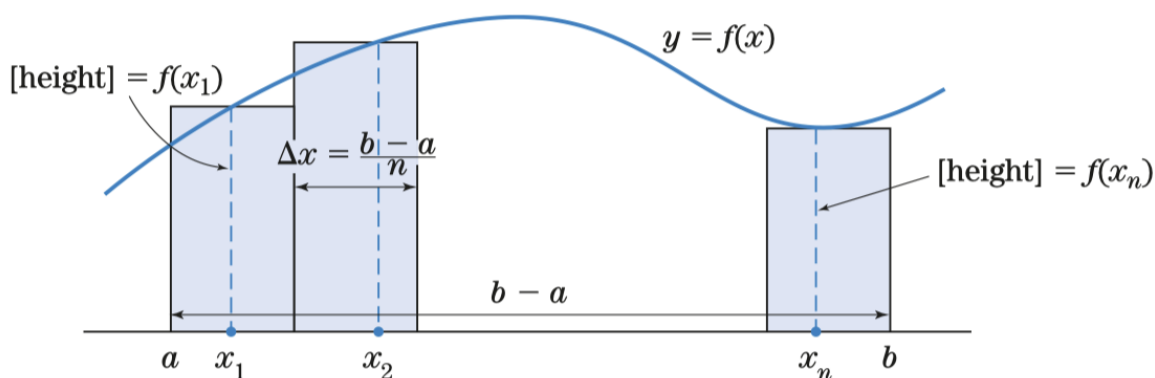


If $f(x)$ is a continuous nonnegative function on the interval $a \leq x \leq b$, then the area under the graph of $f(x)$, above the x -axis, from $x = a$ to $x = b$ is equal to the definite integral of f from a to b :

$$\left[\begin{array}{l} \text{Area under the graph of } f \\ \text{from } x = a \text{ to } x = b \end{array} \right] = \int_a^b f(x) dx = F(b) - F(a),$$

where F is an antiderivative of f .

$$F'(x) = f(x)$$



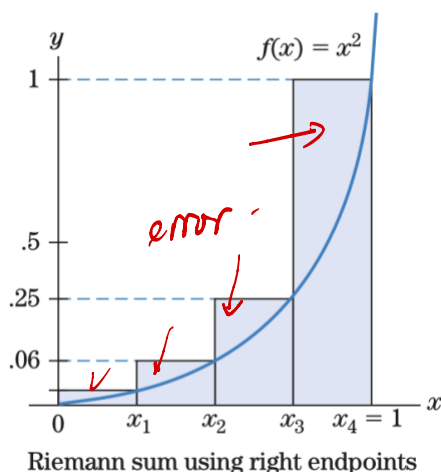
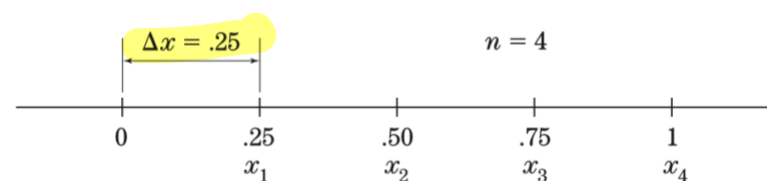
Riemann sum.

$$f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_n)\Delta x.$$

$$\left[\begin{array}{l} \text{Area under the graph} \\ \text{of } f \text{ from } a \text{ to } b \end{array} \right] = \lim_{\Delta x \rightarrow 0} [f(x_1) + f(x_2) + \cdots + f(x_n)]\Delta x.$$

EXAMPLE 9 A Riemann Sum Approximation of an Area

Use a Riemann sum with $n = 4$ to estimate the area under the graph of $f(x) = x^2$ from 0 to 1. Select the right endpoints of the subintervals as x_1, x_2, x_3, x_4 .



$$A = \int_0^1 x^2 dx = \frac{x^3}{3} \Big|_0^1 = \frac{1}{3} \approx 0.33$$

SOLUTION

Here, $\Delta x = (1 - 0)/4 = .25$. The right endpoint of the first subinterval is $0 + \Delta x = .25$. We obtain subsequent right endpoints by successively adding .25.

The right endpoints here give an area estimate that is obviously greater than the exact area. Midpoints would work better. But if the rectangles are sufficiently narrow, even a Riemann sum using right endpoints will be close to the exact area.

The corresponding Riemann sum is

$$\begin{aligned} A &= f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x + f(x_4)\Delta x \\ &= [f(x_1) + f(x_2) + f(x_3) + f(x_4)]\Delta x \\ &= [(.25)^2 + (.5)^2 + (.75)^2 + (1)^2](.25) \\ &= [.0625 + .25 + .5625 + 1](.25) \\ &= 1.875 \cdot (.25) = .46875. \end{aligned}$$

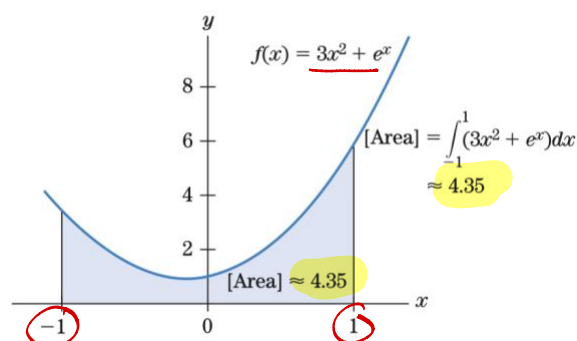
Fundamental Theorem of Calculus

Let f be a continuous function on the interval $[a, b]$, with an antiderivative F . Then, the Riemann sums approach the definite integral of f on $[a, b]$, as the number of subintervals increases indefinitely. That is,

$$\lim_{\Delta x \rightarrow 0} [f(x_1) + f(x_2) + \cdots + f(x_n)]\Delta x = \int_a^b f(x) dx = F(b) - F(a).$$

EXAMPLE 10 Area under the Graph of a Function

Compute the area of the shaded region under the graph of $f(x) = 3x^2 + e^x$, from $x = -1$ to $x = 1$.



$$\int 3x^2 dx = 3 \frac{x^{2+1}}{2+1} = \frac{3}{3} x^3 = x^3$$

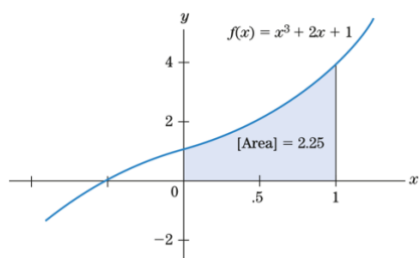
SOLUTION

$f(x)$ is a nonnegative function. The area under the graph, from -1 to 1 , is

$$\begin{aligned} A &= \int_{-1}^1 (3x^2 + e^x) dx = (x^3 + e^x) \Big|_{-1}^1 = (1^3 + e^1) - (-1^3 + e^{-1}) \\ &= 2 + e - 1/e \approx \underline{4.35}. \end{aligned}$$

EXAMPLE 11 Area under a Cubic

Compute the area of the shaded region under the graph of $f(x) = x^3 + 2x + 1$, from $x = 0$ to $x = 1$.



SOLUTION

$$\begin{aligned} A &= \int_0^1 (x^3 + 2x + 1) dx = \left(\frac{x^4}{4} + 2 \frac{x^2}{2} + x \right) \Big|_0^1 \\ &= \left[\frac{(1)^4}{4} + (1)^2 + 1 \right] - 0 \\ &= \frac{1}{4} + 1 + 1 \\ &= \underline{2.25} \end{aligned}$$

Calculator →

The average value (mean value) of a continuous function

The average value of a continuous function $f(x)$ over the interval $a \leq x \leq b$ is defined as the quantity

$$\frac{1}{b-a} \int_a^b f(x) dx.$$

EXAMPLE 12 Average Value (Mean Value)

Compute the average value of $f(x) = \sqrt{x}$ over the interval $0 \leq x \leq 9$.

SOLUTION

$a = 0$ and $b = 9$. We find the average value of $f(x) = \sqrt{x}$ over the interval $0 \leq x \leq 9$ is equal to

$$\frac{1}{9-0} \int_0^9 \sqrt{x} dx$$

$$\int x^{\frac{1}{2}} dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1}$$

Since $\sqrt{x} = x^{1/2}$, an antiderivative of $f(x) = \sqrt{x}$ is $F(x) = \frac{2}{3}x^{3/2}$. Therefore,

$$\frac{1}{9} \int_0^9 \sqrt{x} dx = \frac{1}{9} \left(\frac{2}{3} x^{3/2} \right) \Big|_0^9 = \frac{1}{9} \left(\frac{2}{3} \cdot 9^{3/2} - 0 \right) = \frac{1}{9} \left(\frac{2}{3} \cdot 27 \right) = 2,$$

$$= \frac{x^{\frac{3}{2}}}{(\frac{3}{2})} = \frac{2}{3} x^{3/2}$$

so that the average value of $f(x) = \sqrt{x}$ over the interval $0 \leq x \leq 9$ is 2.

(mean value)

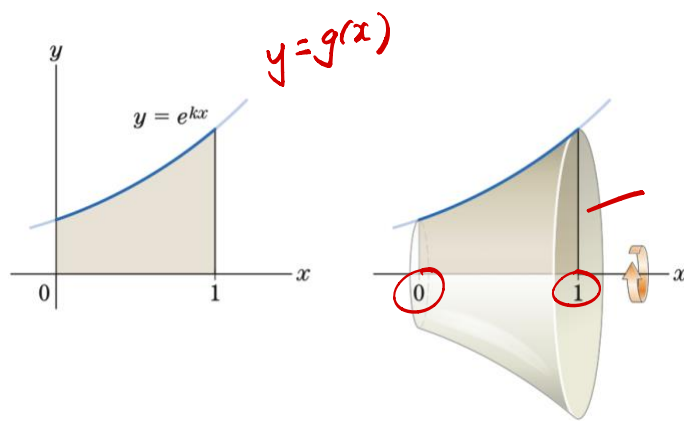
The volume of the solid of revolution

The volume of the solid of revolution obtained from revolving the region below the graph of $y = g(x)$ from $x = a$ to $x = b$ about the x -axis is

$$V = \int_a^b \pi [g(x)]^2 dx \quad \checkmark$$

EXAMPLE 13 Volume of a Solid

Find the volume of the solid of revolution we obtain by revolving the region of Fig. 6 about the x -axis.



$$\int e^{kx} dx = \frac{e^{kx}}{k}$$

SOLUTION

Here, $g(x) = e^{kx}$, and

$$\begin{aligned}
 V &= \underline{\text{[volume]}} = \int_0^1 \pi (e^{kx})^2 dx = \int_0^1 \pi e^{2kx} dx = \frac{\pi}{2k} e^{2kx} \Big|_0^1 = \frac{\pi}{2k} (e^{2k} - 1) \quad \checkmark \\
 &= \frac{\pi}{2k} (e^{2k} - e^0) \\
 e^0 &= 1
 \end{aligned}$$

EXAMPLE 14 Volume of a Cone

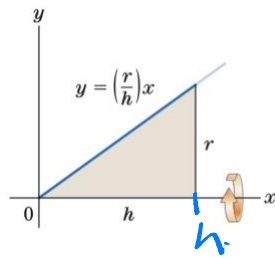
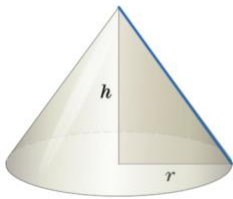
Find the volume of a right circular cone of radius r and height h .

SOLUTION

$$y = g(x) = \left(\frac{r}{h}\right)x$$

$$\int x^2 dx = \frac{x^3}{3}$$

$$\begin{aligned} \int_0^h \pi \left(\frac{r}{h}x\right)^2 dx &= \frac{\pi r^2}{h^2} \int_0^h x^2 dx = \frac{\pi r^2}{h^2} \left[\frac{x^3}{3}\right]_0^h = \frac{\pi r^2 h}{3} \\ &= \frac{\pi r^2}{h^2} \frac{h^3}{3} \end{aligned}$$

**The Trigonometric functions****Radian Measure of Angles**

To relate degrees and radians, we can use

$$d^\circ = d \times \frac{\pi}{180} \text{ radians}$$

$$\pi \text{ rad} = 180^\circ$$

In particular,

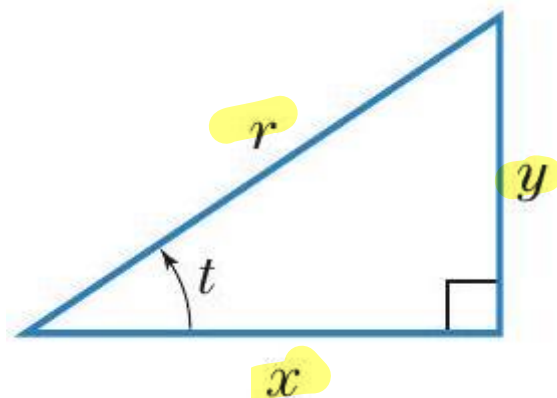
$$360^\circ = 2\pi \text{ radians.}$$

An angle is in standard position on a coordinate system when the vertex of the angle is at $(0,0)$ and one side, called the initial side, is along the positive x -axis. We measure such an angle from the initial side to the terminal side, where a counterclockwise angle is positive and a clockwise angle is negative.

The Sine and the Cosine

Let t represent the measure in radians of an angle as in the figure. We have

$$\sin t = \frac{y}{r} \text{ and } \cos t = \frac{x}{r}$$



The following are some of the important properties of the sine and cosine functions

$$\cos^2 t + \sin^2 t = 1$$

$$\cos(t \pm 2\pi) = \cos t, \sin(t \pm 2\pi) = \sin t$$

$$\cos(-t) = \cos t, \sin(-t) = -\sin t$$

$$\sin(s + t) = \sin s \cos t + \cos s \sin t$$

EXAMPLES

1. Let t be an angle between 0° and 90° as in the figure. Determine $\sin t$ and $\cos t$.

Solution In the right triangle, the hypotenuse is equal to

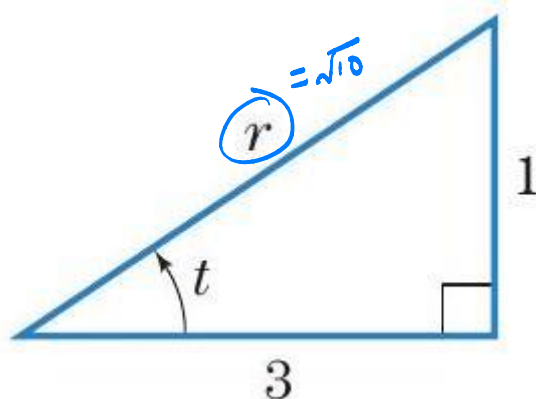
$$r = \sqrt{1^2 + 3^2} = \sqrt{10}$$

So,

$$\sin t = \frac{1}{\sqrt{10}} \text{ and } \cos t = \frac{3}{\sqrt{10}}$$

$$r^2 = 1^2 + 3^2$$

$$r = \sqrt{1^2 + 3^2}$$



2. Given $\sin t = .3$, find $\cos t$

(a) if t is an angle in the first quadrant.

(b) if t is an angle in the second quadrant.

Solution (a) Using $\cos^2 t + \sin^2 t = 1$, we get

$$\begin{aligned}\cos^2 t + (.3)^2 &= 1 \\ \cos^2 t &= 1 - .09 = .91 \\ \cos t &= \pm\sqrt{.91} \approx \pm.954\end{aligned}$$



If the angle is in the first quadrant, $\sin t$ and $\cos t$ are positive. In this case, $\cos t = \sqrt{.91} \approx .954$. (b) If the angle is in the second quadrant, $\sin t$ is positive while $\cos t$ is negative. In this case, $\cos t = -\sqrt{.91} \approx -.954$.

Differentiation and Integration of $\sin t$ and $\cos t$

The two basic differentiation formulas are the following.

$$\frac{d}{dt} \sin t = \cos t, \quad \frac{d}{dt} \cos t = -\sin t$$

Using the chain rule, we generalize these formulas as follows.

$$\begin{aligned}\bullet \quad \frac{d}{dt} [\sin g(t)] &= \cos g(t) \frac{d}{dt} [g(t)] \\ \frac{d}{dt} [\cos g(t)] &= -\sin g(t) \frac{d}{dt} [g(t)]\end{aligned}$$

Reversing the differentiation formulas, we obtain the following integration formulas.

$$\int \cos t \, dt = \sin t + C \quad \text{and} \quad \int \sin t \, dt = -\cos t + C$$

More generally, if $a \neq 0$,

$$\int \cos(at) \, dt = \frac{1}{a} \sin(at) + C$$

$$\frac{d}{dt} \left(\frac{1}{a} \sin(at) \right) = \left(\frac{1}{a} \right) a \cos(at) = \cos at$$

and

$$\int \sin(at) \, dt = -\frac{1}{a} \cos(at) + C$$

$$\begin{aligned}\frac{d}{dt} \left(-\frac{1}{a} \cos(at) \right) &= \left(-\frac{1}{a} \right) (-a) \sin(at) \\ &= \sin at\end{aligned}$$

EXAMPLE 15 Derivatives with the Sine Function

Differentiate

(a) $\sin 3t$

(b) $(t^2 + 3\sin t)^5$.

SOLUTION

$$(a) \quad \frac{d}{dt} (\sin 3t) = (\cos 3t) \frac{d}{dt} (3t) = (\cos 3t) \cdot 3 = 3 \cos 3t$$

$$\sin 3t + C = \int 3 \cos 3t \, dt$$

$$\begin{aligned}
 (b) \quad & \frac{d}{dt} (t^2 + 3 \sin t)^5 \\
 &= 5(t^2 + 3 \sin t)^4 \left[\frac{d}{dt}(t^2) + 3 \frac{d}{dt}(\sin t) \right] \\
 &= 5(t^2 + 3 \sin t)^4 [2t + 3 \cos t]
 \end{aligned}$$

EXAMPLE 16 Derivatives with the Cosine Function

Differentiate

(a) $\cos(t^2 + 1)$

(b) $\cos^2 t$

SOLUTION

(a) $\frac{d}{dt} \cos(t^2 + 1) = -\sin(t^2 + 1) \frac{d}{dt}(t^2 + 1) = -\sin(t^2 + 1) \cdot (2t)$ Use (4).

$$= -2t \sin(t^2 + 1) \quad \checkmark$$

(b)

$$\begin{aligned}
 \frac{d}{dt} \cos^2 t &= \frac{d}{dt} (\cos t)^2 \\
 &= 2 \cos t \frac{d}{dt} (\cos t) \\
 &= 2 \cos t (-\sin t) \\
 &= -2 \cos t \sin t \quad \times
 \end{aligned}$$

EXAMPLE 17 Differentiating Sine and Cosine Functions

Differentiate

(a) $t^2 \cos 3t$

(b) $(\sin 2t)/t$.

SOLUTION

(a) From the product rule we have

$$\begin{aligned} \frac{d}{dt}(t^2 \cos 3t) &= t^2 \frac{d}{dt} \cos 3t + (\cos 3t) \frac{d}{dt} t^2 \quad \text{Product rule.} \\ &= t^2(-3\sin 3t) + (\cos 3t)(2t) \quad \text{Use (4).} \\ &= -3t^2 \sin 3t + 2t \cos 3t. \end{aligned}$$

(b) From the quotient rule we have

$$\frac{d}{dt} \left(\frac{\sin 2t}{t} \right) = \frac{t \frac{d}{dt} \sin 2t - (\sin 2t) \cdot 1}{t^2} = \frac{2t \cos 2t - \sin 2t}{t^2}.$$

EXAMPLE 18 Integrals of Sine Functions

Calculate the following indefinite integrals.

(a) $\int \sin t \, dt$

(b) $\int \sin 3t \, dt$

(a) Since $\frac{d}{dt}(-\cos t) = \sin t$, we have

$$\int \sin t \, dt = -\cos t + C$$

where C is an arbitrary constant.(b) From part (a) we guess that an antiderivative of $\sin 3t$ should resemble the function $-\cos 3t$. However, if we differentiate, we find that

$$\frac{d}{dt}(-\cos 3t) = (\sin 3t) \cdot \frac{d}{dt}(3t) = 3\sin 3t,$$

which is three times too much. So we multiply this last equation by $\frac{1}{3}$ to derive that

$$\frac{d}{dt} \left(-\frac{1}{3} \cos 3t \right) = \sin 3t,$$

so

$$\int \sin 3t \, dt = -\frac{1}{3} \cos 3t + C$$

$$\int \sin 3t \, dt = \frac{-\cos 3t}{3} + C$$

The Tangent and Other Trigonometric Functions

Other basic trigonometric functions involving the sine and the cosine are defined as follows.

$$\tan t = \frac{\sin t}{\cos t}; \cot t = \frac{\cos t}{\sin t}$$

$$\sec t = \frac{1}{\cos t}; \csc t = \frac{1}{\sin t}$$

We obtain the derivatives of these functions by using the derivative rules for the sine and the cosine and various differentiation formulas, such as the quotient rule. We have

$$\frac{d}{dt} \tan t = \sec^2 t; \frac{d}{dt} \cot t = -\csc^2 t$$

Reversing these formulas, we obtain

$$\int \sec^2 t dt = \tan t + C; \int \csc^2 t dt = -\cot t + C$$

More generally, if $a \neq 0$,

$$\int \sec^2 at dt = \frac{1}{a} \tan at + C$$

$$\int \csc^2 at dt = -\frac{1}{a} \cot at + C$$

EXAMPLES

Differentiate

(a) $\tan(t^2 + 2t)$

(b) $t \tan^2 t$.

Solution

$$(a) \frac{d}{dt} [\tan(t^2 + 2t)] = \sec^2(t^2 + 2t) \frac{d}{dt} [t^2 + 2t]$$

$$= (2t + 2) \sec^2(t^2 + 2t)$$

$$(b) \frac{d}{dt} [t \tan^2 t] = t \frac{d}{dt} [\tan^2 t] + \tan^2 t \left(\frac{d}{dt} t \right) = 1$$

$$= t(2 \tan t) \frac{d}{dt} [\tan t] + \tan^2 t$$

$$= 2t \tan t \sec^2 t + \tan^2 t \quad \times$$

EXAMPLES

Evaluate the integrals

(a) $\int \sec^2 5t dt$

(b) $\int \frac{4}{\cos^2 3t} dt$

Solution

(a) $\int \sec^2 5t dt = \frac{1}{5} \tan 5t + C$

(b) $\int \frac{4}{\cos^2 3t} dt = \int 4 \sec^2 3t dt = \frac{4}{3} \tan 3t + C$

A summary of Differentiation Formula and Corresponding Integration Formula

Differentiation Formula	Corresponding Integration Formula
$\frac{d}{dx}(x^r) = rx^{r-1}$	$\int rx^{r-1} dx = x^r + C$ or
$\frac{d}{dx}\left(\frac{x^{r+1}}{r+1}\right) = x^r$	$\int x^r dx = \frac{x^{r+1}}{r+1} + C, r \neq -1$
$\frac{d}{dx}(e^x) = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx}(\ln x) = \frac{1}{x}$	$\int \frac{1}{x} dx = \ln x + C$
$\frac{d}{dx}(\sin x) = \cos x$	$\int \cos x dx = \sin x + C$
$\frac{d}{dx}(\cos x) = -\sin x$	$\int \sin x dx = -\cos x + C$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\int \sec^2 x dx = \tan x + C$

$\frac{d}{dx}(e^{ax}) = ae^{ax}$
 $\frac{d}{dx}(\ln(ax)) = \frac{1}{x}$

$\int e^{ax} dx = \frac{e^{ax}}{a} + C$
 $\int \frac{1}{ax} dx = \frac{1}{a} \ln |ax| + C$

Integration by Substitution

Every differentiation formula can be turned into a corresponding integration formula. If we apply this reasoning to the chain rule, we obtain the integration by substitution formula:

$$\int f(g(x))g'(x)dx = F(g(x)) + C$$

where C is any constant and F is an antiderivative of f . This formula suggests a technique for evaluating integrals of products of functions by using a substitution $u = g(x)$ and $du = g'(x)dx$. This substitution will turn an integral of the form

$$\int f(g(x))g'(x)dx$$

into an integral that is easier to evaluate, since

$$\int f(u)du = F(u) + C$$

The success of the method of integration by substitution depends on our choice of the substitution. Since there is no rule for selecting the "right" substitution, you can improve your skill in this area by considering as many examples as possible.

EXAMPLE 19 Integration by Substitution

Determine

$$\int (x^2 + 1)^3 \cdot 2x dx$$

FIRST SOLUTION

If we set $f(x) = x^3$, $g(x) = x^2 + 1$, then $f(g(x)) = (x^2 + 1)^3$ and $g'(x) = 2x$. Therefore, we can apply formula (1). An antiderivative $F(x)$ of $f(x)$ is given by

$$F(x) = \frac{1}{4}x^4,$$

so that, by formula (1), we have

$$\int (x^2 + 1)^3 \cdot 2x dx = F(g(x)) + C = \frac{1}{4}(x^2 + 1)^4 + C.$$

Let $u = x^2 + 1$ $\frac{du}{dx} = 2x$
 $du = 2x dx$

$$\int (x^2 + 1)^3 \cdot 2x dx = \int u^3 du$$

$$= \frac{u^4}{4} + C$$

$$= \frac{(x^2 + 1)^4}{4} + C$$

Check $\frac{d}{dx} \left(\frac{(x^2 + 1)^4}{4} + C \right) = \frac{4(x^2 + 1)^3 (2x)}{4} + 0$

$$= (x^2 + 1)^3 2x \quad \checkmark$$

EXAMPLE 20 Substitution with an ExponentialEvaluate $\int 2xe^{x^2} dx = \int e^{x^2} 2x dx$ SOLUTION Let $u = x^2$, so $du = \frac{d}{dx}(x^2)dx = 2x dx$. Therefore,

$$\frac{du}{dx} = 2x$$

$$\begin{aligned} \int 2xe^{x^2} dx &= \int e^{x^2} \cdot 2x dx \\ &= \int e^u du \\ &= e^u + C \\ &= e^{x^2} + C \end{aligned}$$

check

$$\begin{aligned} \frac{d}{dx}(e^{x^2} + C) \\ &= e^{x^2} (2x) + 0 \\ &= 2x e^{x^2} \checkmark \end{aligned}$$

Integration by Substitution

- 1 Define a new variable $u = g(x)$, where $g(x)$ is chosen in such a way that, when written in terms of u , the integrand is simpler than when written in terms of x .
- 2 Transform the integral with respect to x into an integral with respect to u by replacing $g(x)$ everywhere by u and $g'(x)dx$ by du .
- 3 Integrate the resulting function of u .
- 4 Rewrite the answer in terms of x by replacing u by $g(x)$.

Let us try a few more examples.

EXAMPLE 21 Substitution with a Radical [7]Evaluate $\int 3x^2 \sqrt{x^3 + 1} dx = \int (x^3 + 1)^{\frac{1}{2}} \cdot 3x^2 dx$

SOLUTION

$$\text{Let } u = x^3 + 1 \quad \frac{du}{dx} = 3x^2 \quad du = 3x^2 dx$$

$$\begin{aligned} \int (x^3 + 1)^{\frac{1}{2}} 3x^2 dx &= \int u^{\frac{1}{2}} du \\ &= \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C \\ &= \frac{2}{3} u^{\frac{3}{2}} + C \\ &= \frac{2}{3} (x^3 + 1)^{\frac{3}{2}} + C \checkmark \end{aligned}$$

check

$$\frac{d}{dx} \left(\frac{2}{3} (x^3 + 1)^{\frac{3}{2}} + C \right) = (x^3 + 1)^{\frac{1}{2}} (3x^2) \checkmark$$

EXAMPLE 22 Substitution with a LogarithmFind $\int \frac{(\ln x)^2}{x} dx$.

$$\frac{du}{dx} = \frac{1}{x}$$

SOLUTION Let $u = \ln x$. Then, $du = (1/x)dx$ and

$$\begin{aligned}
 \int \frac{(\ln x)^2}{x} dx &= \int (\ln x)^2 \cdot \frac{1}{x} dx \\
 &= \int u^2 du \\
 &= \frac{u^3}{3} + C \\
 &= \frac{(\ln x)^3}{3} + C. \quad \text{Since } u = \ln x.
 \end{aligned}$$

EXAMPLE 23 Substitution with an ExponentialFind $\int x^2 e^{x^3} dx = \int e^{x^3} x^2 dx$

SOLUTION

$$\text{Let } u = x^3 \quad \frac{du}{dx} = 3x^2 \quad \frac{du}{3} = x^2 dx$$

$$\begin{aligned}
 \int e^{x^3} x^2 dx &= \int e^u \frac{du}{3} \\
 &= \frac{1}{3} \int e^u du \\
 &= \frac{1}{3} e^u + C \\
 &= \frac{1}{3} e^{x^3} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(Check)} \quad \frac{d}{dx} \left(\frac{1}{3} e^{x^3} + C \right) &= \frac{1}{3} e^{x^3} (3x^2) + 0 \\
 &= \frac{1}{3} e^{x^3} x^2
 \end{aligned}$$

EXAMPLES

Evaluate the integrals by substitution:

(a) $\int 3x^2(x^3 + 7)dx$

(b) $\int \frac{x}{(x^2+1)^3} dx$

(c) $\int \frac{1}{x \ln x} dx$

(d) $\int e^x \sqrt{e^x + 1} dx$

Solution

(a) Let $u = x^3 + 7$, $du = 3x^2 dx$. Then,

$$\frac{du}{dx} = 3x^2$$

$$\int 3x^2(x^3 + 7)dx = \int u du = \frac{1}{2}u^2 + C$$

$$= \frac{1}{2}(x^3 + 7)^2 + C$$

(b) Let $u = x^2 + 1$, $du = 2x dx$. Since the integral does not contain the expression $2x dx$, we will use the equivalent substitution $x dx = \frac{1}{2} du$. Then,

$$\int \frac{x}{(x^2 + 1)^3} dx = \int \frac{\frac{1}{2} du}{u^3} = \frac{1}{2} \int u^{-3} du$$

$$= \frac{1}{2} \left(\frac{1}{-2} \right) u^{-2} + C$$

Since $\int u^r du = \frac{1}{r+1} u^{r+1} + C (r \neq -1)$

$$= \frac{1}{4} \frac{1}{u^2} + C = -\frac{1}{4} \frac{1}{(x^2 + 1)^2} + C$$

$$(c) \int \frac{1}{x \ln x} dx = \int \frac{1}{\ln x} \cdot \frac{1}{x} dx$$

Let $u = \ln x$

$$\frac{du}{dx} = \frac{1}{x}$$

$$du = \frac{1}{x} dx$$

$$= \int \frac{1}{u} du$$

$$= \ln|u| + C$$

$$= \ln|\ln x| + C$$

$$(d) \int e^x \sqrt{e^x + 1} dx = \int (e^x + 1)^{\frac{1}{2}} e^x dx$$

Let $u = e^x + 1$

$$\frac{du}{dx} = e^x$$

$$du = e^x dx$$

$$= \int u^{\frac{1}{2}} du$$

$$= \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C = \frac{2}{3} u^{\frac{3}{2}} + C$$

$$= \frac{2}{3} (e^x + 1)^{3/2} + C$$

Integration by Parts

Integration by parts is an important integration technique based on reversing the product rule differentiation formula. Let $f(x)$ and $g(x)$ be any two functions and let $G(x)$ be an antiderivative of $g(x)$. The product rule asserts that

$$\begin{aligned}\frac{d}{dx}[f(x)G(x)] &= f(x)G'(x) + f'(x)G(x) \\ &= f(x)g(x) + f'(x)G(x)\end{aligned}$$

$$G(x) = \int g(x) dx$$

Since $G'(x) = g(x)$.

Therefore,

$$f(x)G(x) = \int f(x)g(x)dx + \int f'(x)G(x)dx$$

This last formula can be rewritten in the following more useful form:

$$\int f(x)g(x)dx = f(x)G(x) - \int f'(x)G(x)dx$$

which is the basis for the integration by parts technique.

EXAMPLES

Evaluate the integrals by using integration by parts:

- (a) $\int x(x+7)^4 dx$
- (b) $\int (x+1)e^{x/5} dx$
- (c) $\int (x^3+x)\ln x dx$

Solution

(a) Clearly the integral would be easier to evaluate if it were not for the presence of the factor x in the integrand. You can get rid of this factor by using integration by parts and setting $f(x) = x$. Then,

$f'(x) = 1$. To complete the setup for integration by parts, we take $g(x) = (x+7)^4$, and so, $G(x) = \frac{1}{5}(x+7)^5$. Applying the integration by parts formula, we find

$$\begin{aligned}\int \overbrace{x \cdot (x+7)^4}^{f(x) \cdot g(x)} dx &= \overbrace{x \cdot \frac{1}{5}(x+7)^5}^{f(x) \cdot G(x)} - \int \overbrace{\left(1 \cdot \frac{1}{5}(x+7)^5\right)}^{f'(x) \cdot G(x)} dx \\ &= \frac{x}{5}(x+7)^5 - \int \frac{1}{5}(x+7)^5 dx \\ &= \frac{x}{5}(x+7)^5 - \frac{1}{5} \left(\frac{1}{6} \right) (x+7)^6 + C \\ &= \frac{x}{5}(x+7)^5 - \frac{1}{30}(x+7)^6 + C\end{aligned}$$

$$\begin{aligned}G(x) &= \int g(x) dx \\ &= \int (x+7)^4 dx \\ &= \frac{(x+7)^5}{5}\end{aligned}$$

(b) This is a situation similar to that in (a), where we can simplify the integral by getting rid of the factor $(x + 1)$ in the integrand. Take $f(x) = x + 1$, $g(x) = e^{x/5}$. Then, $f'(x) = 1$ and $G(x) = 5e^{x/5}$. (In computing $G(x)$, we used the formula

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, a \neq 0$$

with $a = 1/5$.) So,

$$\begin{aligned} \int \overbrace{(x+1)}^{f(x)} \cdot \overbrace{e^{x/5}}^{g(x)} dx &= \overbrace{(x+1)}^{f(x)} \cdot \overbrace{5e^{x/5}}^{G(x)} - \int \overbrace{(1 \cdot 5e^{x/5})}^{f'(x) \cdot G(x)} dx \\ &= 5(x+1)e^{x/5} - 5 \int e^{x/5} dx \\ &= 5(x+1)e^{x/5} - 25e^{x/5} + C \\ &= (5x - 20)e^{x/5} + C. \end{aligned}$$

$$\begin{aligned} G(x) &= \int g(x) dx \\ &= \int e^{x/5} dx \\ &= \frac{e^{x/5}}{(1/5)} + C \\ &= 5e^{x/5} + C \end{aligned}$$

(c) Take $f(x) = \ln x$, $g(x) = x^3 + x$. So, $f'(x) = \frac{1}{x}$ and $G(x) = \frac{1}{4}x^4 + \frac{1}{2}x^2$. Then,

$$\begin{aligned} \int \overbrace{(x^3 + x)}^{f(x)} \ln x \, dx &= \overbrace{\ln x \left(\frac{1}{4}x^4 + \frac{1}{2}x^2 \right)}^{f(x) \cdot G(x)} - \int \overbrace{\frac{1}{x} \cdot \left(\frac{1}{4}x^4 + \frac{1}{2}x^2 \right)}^{f'(x) \cdot G(x)} dx \\ &= \frac{x^2}{2} \left(\frac{1}{2}x^2 + 1 \right) \ln x - \int \left(\frac{1}{4}x^3 + \frac{1}{2}x \right) dx \\ &= \frac{x^2}{2} \left(\frac{1}{2}x^2 + 1 \right) \ln x - \frac{1}{16}x^4 - \frac{1}{4}x^2 + C \end{aligned}$$

$$\begin{aligned} G(x) &= \int g(x) dx \\ &= \int x^3 + x \, dx \\ &= \frac{x^4}{4} + \frac{x^2}{2} \end{aligned}$$

